

Comparison of the cytotoxic and mutagenic potential of liquid smoke food flavourings, cigarette smoke condensate and wood smoke condensate.



Although products of pyrolysis are often cytotoxic and mutagenic, the relationship between the type of material pyrolysed and the toxicity of the resulting pyrolysis products is poorly understood. The objective of this study was to evaluate and compare the cytotoxicity and mutagenicity of several types of common pyrolysis products. The cytotoxicity and mutagenicity of these products were assessed by using neutral red uptake and Ames mutagenicity assays, respectively. The biological activities of four liquid smoke food flavourings (LSF) were compared with two other pyrolysis-derived materials; cigarette smoke condensate (CSC) and a wood smoke condensate (WSC). Results indicated all of the mixtures exhibited a concentration-dependent cytotoxic response. The CSC and WSC were less cytotoxic than three of the LSFs, but more cytotoxic than one of the brands. The CSC was mutagenic in two *Salmonella* strains; however, none of the LSFs or WSC was mutagenic using TA98, and only three of the LSFs were positive with TA100. The six pyrolysis-derived materials evaluated in this study showed differing patterns and magnitudes of cytotoxicity and mutagenicity. These results indicate that the cytotoxicity and mutagenicity of complex mixtures derived from pyrolysis products are affected by the type of material pyrolysed and/or the method used to prepare the mixture. The cytotoxic potential of some commercial smoke flavourings is greater than cigarette smoke condensate and several of the food flavourings are mutagenic in one *Salmonella* strain.